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# Shunxin Wang

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## Research Interests

• Robust Computer Vision • Generalization • Efficient Learning

## Work Experiences

**PhD Candidate in Artificial Intelligence**, University of Twente, The Netherlands, Nov. 2020 — Exp. Oct. 2024

- Topic: **Efficient Learning for Robust Computer Vision**
  Analyze the behavior of Convolutional Neural Networks in image classification from a frequency shortcut perspective to develop an efficient learning algorithm while guaranteeing the robustness of models.
- Project: **Robust Computer Vision Systems** for cell analysis
  Improve image quality and implement cell detection and segmentation with colleagues from BIOS (biomedical application systems) Lab at University of Twente.

## Education

(Ongoing) PhD in Artificial Intelligence, University of Twente, Nov. 2020 — Exp. Oct. 2024

MSc in Computer Science (Computer Vision and Biometrics), University of Twente, Sep. 2019 — Aug. 2020

BSc in Electronics and Information Engineering, Northwestern Polytechnical University, Sep. 2015 — Jul. 2019

## Activities and awards

Student volunteer | Women in Computer Vision Workshop at ICCV 2023

Best Abstract Award | Nederlandse Vereniging voor Patroonherkenning en Beeldverwerking (NVPHBV) Spring meeting on Robust AI 2023

Participant | The $10^{th}$ Vision Understanding and Machine Intelligence Summer School 2022

Scholarship recipient | University Twente Scholarship 2019

Scholarship recipient | Wu Yajun Scholarship 2018

## Publications and Preprints

• ***S. Wang**, R. Veldhuis, C. Brune, and N. Strisciuglio*, "What Do Neural Networks Learn in Image Classification? A Frequency Shortcut Perspective", **ICCV** 2023.

• ***S. Wang**, C. Brune, R. Veldhuis, and N. Strisciuglio*, "DFM-X: Augmentation by Leveraging Prior Knowledge of Shortcut Learning", **ICCVW VIPriors** 2023 , Paris, France.

• ***S. Wang**, R. Veldhuis, and N. Strisciuglio*, "Larger is not better: A Survey on the Robustness of Computer Vision Models against Common Corruptions". *Under review*

• *I. Mazilu, **S. Wang**, S. Dummer, R. Veldhuis, C. Brune, and N. Strisciuglio*, "Defocus Blur Synthesis and Deblurring via Interpolation and Extrapolation in Latent Space", **CAIP** 2023.

• ***S. Wang**, R. Veldhuis, C. Brune, and N. Strisciuglio*, "Frequency Shortcut Learning in Neural Networks," in **NeurIPS** Workshop on Distribution Shifts: Connecting Methods and Applications, 2022.

• ***S. Wang**, U. M. Kelly, and R. N. J. Veldhuis*, "Gender Obfuscation through Face Morphing," 2021 IEEE International Workshop on Biometrics and Forensics (IWBF), 2021.

## Skills

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|---|---|
| **Programming Languages \| Libraries:** | Python, Matlab, C++ \| Torch, Scikit-learning, Pandas, Numpy, Lightning, SciPy (also experience with Keras, Tensorflow) |
| **Languages** | English (Working proficiency), Mandarin (Native) |